RESEARCH ARTICLE

Type 1 Diabetes and Incident Dementia

An Analysis in the All of Us Cohort

Anna M. Pederson,[1,2] Peter Buto,[1,2] Scott C. Zimmerman,[1,2] Mabeline Velez,[1] Kendra D. Sims,[1] Audrey R. Murchland,[1] Jingxuan Wang,[1] Alana T. Brennan,[1,3] M. Maria Glymour,[1] and Jennifer Weuve[1]

Correspondence
Ms. Pederson
anniep@bu.edu

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Abstract

Background and Objectives

Although diabetes mellitus (DM) is a well-established determinant of dementia risk, most studies have evaluated type 2 DM (T2DM) or any DM. The influence of type 1 DM (T1DM) on dementia risk remains unclear. We evaluated associations of T1DM and T2DM, separately, with incident dementia using linked electronic health records (EHRs).

Methods

This prospective cohort study used previously collected survey and EHR data from the All of Us (AoU) cohort, a convenience sample of US adults. Eligible participants were 50 years or older and completed baseline surveys. Enrollment began in 2017, with data available through October 2023, along with records before enrollment. Mean follow-up from baseline was 2.4 years. We developed an algorithm to distinguish DM type based on count of T1DM encounters. This algorithm was validated against 2 reference measures: self-reported diabetes type and C-peptide values. Using AoU data, we classified participants as having no DM, T1DM, or T2DM. We ascertained incident dementia using ICD-9, ICD-10, and Systematized Nomenclature of Medicine codes in participants' EHRs. We estimated hazard ratios (HRs) and 95% CIs for the association of diabetes type with incident dementia using Cox proportional hazards models.

Results

Among 283,772 participants (mean [SD] age 64.62 [8.96] years; 56.7% women), 60.3% identified as non-Hispanic White and 13.3% as Hispanic/Latino. Optimal DM classification algorithm cutoffs varied by reference standard: self-reported diabetes: ≥1 T1DM EHR encounter (sensitivity: 0.59; specificity: 0.90) and C-peptide: ≥3 T1DM EHR encounters (sensitivity: 0.76; specificity: 0.79). Defining T1DM as having ≥1 T1DM encounter, 5,442 participants had T1DM. Compared with those without DM, participants with T1DM had higher dementia incidence (sociodemographic-adjusted HR 2.82; 95% CI 2.28–3.48) and those with T2DM also had elevated risk (sociodemographic-adjusted HR 2.08; 95% CI 2.87–2.31). Results were similar across sex, race, and ethnicity strata.

Discussion

In AoU, DM was associated with elevated dementia risk, with the highest risk among those with T1DM. These findings highlight the need to better understand mechanisms tying T1DM to dementia.

Introduction

As the US population ages and dementia prevalence rises, identifying determinants of dementia risk is an urgent public health priority. Diabetes mellitus (DM), a leading cause of morbidity and mortality, has well-documented links to dementia.[1-3] Type 2 DM (T2DM) accounts for approximately 95% of diabetes cases. Although type 1 DM (T1DM) is far less prevalent, life expectancy with this condition has historically been shorter than life expectancy with T2DM.[4-7] This has limited research on the late-life health of persons with T1DM, including research on

[1]Department of Epidemiology, Boston University School of Public Health, MA; [2]Department of Epidemiology and Biostatistics, University of California San Francisco; and [3]Department of Global Health, Boston University School of Public Health, MA.

Glossary

AoU = All of Us; **DM** = diabetes mellitus; **EHR** = electronic health record; **HR** = hazard ratio; **ICD-9** = International Classification of Diseases, Ninth Revision; **ICD-10** = International Classification of Diseases, 10th Revision; **IRB** = Institutional Review Board; **NH** = non-Hispanic; **PAF** = population attributable fraction; **SNOMED** = Systematized Nomenclature of Medicine; **STROBE** = Strengthening of the Reporting of Observational Studies in Epidemiology; **T1DM** = type 1 DM; **T2DM** = type 2 DM.

dementia.[8-12] Advances in medical care have improved survival in T1DM making such studies increasingly relevant and feasible. Understanding the roles of T1DM and T2DM in dementia risk is also integral to reducing racial and ethnic disparities because T2DM is more common among non-Hispanic (NH) Black and Hispanic individuals than among NH White individuals, and quality of care for both T2DM and T1DM is lower among racially and ethnically marginalized groups.[13-16]

Improvements in T1DM treatment have extended many individuals' lives such that they face aging-related diseases, including dementia.[17] Existing studies of the association of T1DM with dementia have notable limitations such as using national registry or hospital-based samples, limited diversity, or a relatively small number of individuals with T1DM.[18-26] Given the increasing number of aging individuals with T1DM, it is crucial to understand the relation of T1DM to dementia risk.

T1DM is rare, and electronic health records (EHRs) offer a critical opportunity to research T1DM and dementia in large samples. However, methods for accurately identifying individuals with T1DM using EHRs are not well developed. In EHR-based cohorts, distinguishing between T1DM and T2DM may be challenging because of diagnostic code overlap and co-occurring T1DM and T2DM.

Using subsamples of a large EHR cohort, we validated an algorithm using 2 reference standard measures for T1DM to identify T1DM cases from EHR encounters. We then examined the association between DM type and dementia incidence, overall and by race and ethnicity, and sex. We hypothesized that, compared with individuals free of DM, those with T1DM would have a higher risk of incident dementia, and elevated risk would differ from that observed among those with T2DM.

Methods

Study Sample

All of Us (AoU) is an ongoing longitudinal cohort of individuals aged 18 years and older recruited as a convenience sample from across the United States.[27] Recruitment began in 2017 through health care provider organizations and volunteer-based enrollment (eTable 1).[28] At enrollment, participants complete baseline surveys on their sociodemographic characteristics and health needs. All participants provide primary consent to join AoU, and they are additionally asked for consent to link to their EHRs; however, providing this consent is optional.[27,28] Thus, some participants contribute only survey data, while others may also contribute EHR data in addition to survey data. Follow-up information for our study was available through October 10, 2023. We followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for cohort studies.[29]

Algorithm for Classifying Diabetes Type

In the absence of uniformly available diagnostic data, we developed a measure to distinguish individuals with T1DM from those with T2DM based on clinical encounter data. EHR encounters were coded based on ICD-9, ICD-10, and corresponding Systematized Nomenclature of Medicine (SNOMED) codes (eTable 2). To evaluate the validity of the EHR-based classification, we leveraged laboratory measures and self-reported diabetes type (eMethods; eTables 3 and 4).

Reference Standards

For identifying persons with T1DM, 2 measures, each available for a subset of the full sample, defined our reference standards: (1) self-reported T1DM or T2DM (whether a participant indicated having T1DM, T2DM, or both on the AoU personal and family history survey) and (2) C-peptide concentration (T1DM: ≤0.20 nmol/L vs T2DM: >0.20 nmol/L). Self-reported information is commonly used in large cohort studies and can offer additional context when clinical diagnostic information data are unavailable. C-peptide laboratory measurements are used to diagnose T1DM and to distinguish it from T2DM because they reflect pancreatic beta cell function.[30] C-peptide measurements were not collected systematically by AoU for all participants but reflect tests ordered in routine clinical care at the discretion of treating clinicians.

Study Sample for Developing the Classification Algorithm

To develop the classification algorithm, we extracted data on all AoU participants, ages older than 18 at enrollment, with an EHR record of diabetes (either T1DM or T2DM) who also had at least 1 reference standard measured: self-reported diabetes (n = 19,718) or C-peptide measurement (n = 973) (eTable 4). Because the availability and size of each reference standard subset differed, the classification algorithm was developed and evaluated separately within each data set.

Statistical Analyses for Developing the Classification Algorithm

For each participant with DM EHR encounters, we separately counted the number of T1DM and T2DM EHR encounters occurring before the baseline survey. We sought to identify a minimum threshold of T1DM visit counts that would best correspond to having T1DM (vs T2DM). The thresholds for the number of T1DM encounters ranged from 0 to 50. To select the optimal threshold and evaluate the performance of the threshold, we used 5-fold nested cross-validation.[31] In this approach, the data were split into 5 outer folds, each serving once as a held out test set. Performance metrics (sensitivity, specificity, and Youden J index) were calculated on these held-out subsets to evaluate how well the threshold would generalize to unseen data, minimizing bias and preventing data leakage. With each outer training set, we further split the data into 4 inner folds. These inner folds were used to evaluate different T1DM encounter thresholds and select the one that maximized the Youden J index for each of the 2 reference standards. The selected threshold was then applied to the outer test set to assess performance.

Among participants with DM, we determined the number of T1DM encounters that maximized the sensitivity and specificity of classifying participants with T1DM (vs T2DM) against each of the 2 reference standards (eFigure 1). We set a threshold of T1DM EHR encounters above which participants were classified as having T1DM and as having T2DM otherwise (irrespective of the number of EHR encounters for T2DM). In other words, participants with a number of T1DM EHR encounters greater than or equal to the threshold were classified as T1DM, and those with fewer encounters were classified as T2DM. Sensitivity and specificity for T1DM were calculated over thresholds ranging from 0 to 50 to systematically evaluate the algorithm's performance and choose an optimal threshold. The optimal threshold for each reference measure was defined as the threshold value that maximized the Youden J index, calculated using the sensitivity and specificity at that threshold.[32,33]

$$J = \max_c\{Se(c) + Sp(c) - 1\}$$

The optimal cutoff, J, is the number of T1DM EHR encounters (c) that maximizes the sum of sensitivity (Se) and specificity (Sp).

Receiver operating characteristic curves were generated using the outer test folds. The predicted probability for T1DM was based on whether a participant met or exceeded the optimal encounter threshold for each reference standard. Threshold specific sensitivity and specificity were overlaid on the receiver operating characteristic curves to illustrate the performance of the classification algorithm across the 2 reference standards (Figure 1).

Estimation of the Association of T1DM on Dementia Incidence

Study Sample

Our analyses included AoU participants who were 50 years or older at enrollment and who had EHR linkage. We excluded 2,126 individuals with a diagnosis of dementia in their EHR before completion of this survey.

Assessment of T1DM and T2DM

Participants with no DM EHR encounter were classified as not having DM. Using our diabetes classification algorithm, we classified those with any diabetes encounter into diabetes types T1DM or T2DM. Individuals were classified as T1DM if they had at least 1 encounter for T1DM (the threshold we deemed most optimal based on our algorithm validation [see Results section *Validation of the Algorithm for Classifying Diabetes Type* and eTables 5 and 6]); those with no T1DM encounter or T1DM encounters below the threshold deemed most optimal, were classified as having T2DM. Participants categorized as both T1DM and T2DM were grouped with the T1DM category because of small sample sizes.

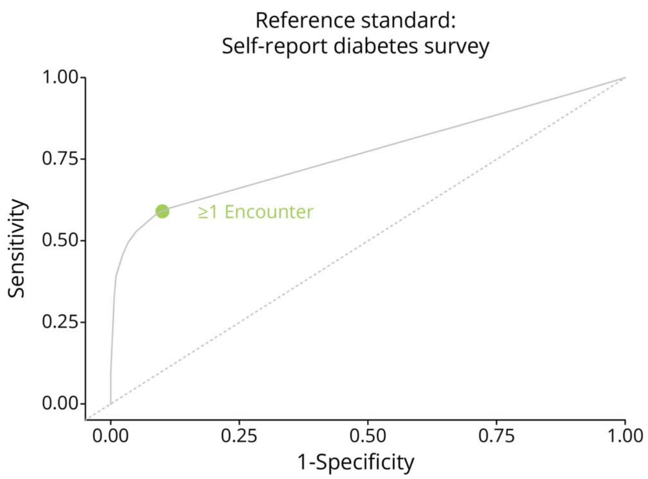
Assessment of All-Cause Dementia

All-cause dementia included Alzheimer dementia, vascular dementia, and dementia of unknown etiology (eTable 7). We did not include diagnosis of frontotemporal, Lewy Body, or alcohol-related dementias because these conditions may preclude the diagnosis of other forms of dementia. We defined time to incident all-cause dementia as time from baseline to the first appearance in a participant's EHR of a corresponding ICD-9, ICD-10, or SNOMED code. Individuals were censored at loss to follow-up, death, or the administrative censoring date defined by the last available EHR encounter date in the sample (October 1, 2023).

Covariates

We estimated the association of each type of DM on dementia incidence, adjusting for 2 sets of covariates representing potential sources of confounding. All covariate data were self-reported at baseline (see eTable 8 for the text of the questions and response options used to collect sex, and race and ethnicity). The first covariate set included sociodemographic factors: age at baseline, sex (man/woman, self-reported in the AoU "The Basics" survey using the "sex" variable; with only responses labeled "male" or "female" included), racial and ethnic identity (NH White, NH Black, NH Asian, Hispanic/Latino, or NH other), educational attainment (self-reported highest level of school completed: less than high school, high school, some college, or college and above), and household income (adjusted by dividing by the square root of household size). The second set additionally included behaviors at midlife that, depending on their timing relative to DM onset, could be sources of confounding or responses to a DM diagnosis. These included smoking history (never smoked or ever smoking at least 100 cigarettes in your life) and alcohol use (never, past, low-use, and high-use, based on the National Institute on Alcohol Abuse and Alcoholism risk categories that account for frequency, quantity, and sex). To avoid overadjusting or adjusting for potential mediators, covariate selection was guided by prior evidence and through the use of directed acyclic graphs that was informed by this evidence (eFigures 2 and 3).

Figure 1 Sensitivity and Specificity of Type 1 Diabetes Classification (vs Type 2 Diabetes) Across 0–50 T1DM EHR Encounter Thresholds, Among All of Us Participants With Diabetes



Performance of T1DM EHR encounter thresholds in predicting T1DM across 2 reference standards: (1) self-report diabetes survey and (2) C-peptide measurements. Each plot shows the pooled sensitivity and specificity across outer test folds for threshold 0 to 50. The sensitivity and specificity at the optimal threshold across the 2 reference standards were as follows: (1) self-report diabetes survey: ≥1 T1DM EHR encounter (sensitivity = 0.59; specificity = 0.90) and (2) C-peptide levels: ≥3 T1DM EHR encounters (sensitivity = 0.76; specificity = 0.79). EHR = electronic health record; T1DM = type 1 diabetes mellitus.

Statistical Analyses

We used Cox proportional hazards regression models, using time from baseline to dementia as the time scale, to estimate hazard ratios (HRs) for the association of DM type, vs no DM, with all-cause dementia. Following baseline, participants with no recorded EHR encounter for 448 days (the 99th percentile of the distribution of days between encounters) were assumed to no longer be receiving care from a clinic with linked EHR data and were censored at 448 days after the date of their last encounter. We fit (1) a model adjusted for sociodemographic factors (set 1) and (2) a model adjusted for both sociodemographic and midlife factors (set 2). We also conducted analyses stratified by race and ethnicity, and sex.

To protect participant privacy, we suppressed results based on fewer than 20 dementia events. In addition, given small participant numbers in some racial and ethnic groups, we combined the NH Asian, NH Black, and NH Middle Eastern and North African subgroups into a single "NH Other" race and ethnicity category. The final categories used in analyses were NH White, Hispanic/Latino, and NH Other.

To quantify the proportion of the dementia incidence among those with T1DM that can be attributed to T1DM, we calculated the attributable fraction among those with T1DM ($AF_{exposed}$), as

$$HR - 1/HR$$

where HR is the hazard ratio for T1DM from the model adjusted for sociodemographic factors (set 1).[34]

To quantify the contribution of T1DM to the incidence of all-cause dementia in our analytical sample, we estimated the following population attributable fraction (PAF), using the Miettinen formula[35,36]:

$$PAF = P_{T1DM} \times (HR - 1)/HR,$$

where P_{T1DM} is the prevalence of T1DM among individuals who developed dementia in our analytical sample and HR is the estimated T1DM-dementia hazard ratio from the model adjusted for sociodemographic factors (set 1).

Sensitivity Analyses

In sensitivity analyses, we defined T1DM as having at least 3 T1DM encounters (instead of at least 1). This definition was deemed most optimal when evaluating the classification algorithm using C-peptide values as the reference standard. For individuals with fewer than 3 T1DM encounters, classification was based on the proportion of their total DM encounters that were for T1DM. If this proportion was greater than or equal to 0.5, they were classified as having T1DM; if this proportion was less than 0.5, they were classified as having T2DM.

In addition, we fit models adjusting for set 2 covariates plus potential factors that could be confounders or mediators including hypertension, dyslipidemia, body mass index, and depression (set 3) (eAppendix 1; eTable 9). Because these factors may lie on causal pathways from T1DM to dementia, the associations generated from these further adjusted analyses should be interpreted with caution.

Standard Protocol Approvals, Registrations, and Patient Consents

This study follows the STROBE guidelines for cohort studies.[29] The AoU Research Program protocol and materials were approved by the program's Institutional Review Board (IRB) before participant enrollment. All participants provided informed consent, and all data made available to researchers through the secure Research Workbench are deidentified and modified to minimize reidentification risk. The Boston University Medical Campus IRB waived approval.

Data Availability

The data used in this study are available through the AoU Researcher Workbench which requires institutional approval and researcher compliance with data use policies. Owing to data use agreements and participant privacy, individual-level data cannot be shared publicly. Code used for the analysis is available on request.

Results

Validation of the Algorithm for Classifying Diabetes Type

Using the Youden J index, the optimal threshold of EHR encounters for classifying T1DM vs T2DM varied across the 2 reference standard measures (Figure 1; eTables 5 and 6). Values reported represent pooled performance across 5 outer folds of the nested cross-validation: ≥1 encounter for self-reported diabetes type (sensitivity: 0.59; specificity: 0.90) and ≥3 encounters for C-peptides (sensitivity: 0.76; specificity: 0.79).

Associations of Diabetes Type With Incident Dementia

Among the 283,772 participants in the analyses of DM and dementia, 5,442 (1.91%) were classified as having T1DM at baseline, based on having at least 1 T1DM EHR encounter, and 51,511 (18.14%) were classified as having T2DM. Over a mean of 2.36 years of follow-up, 2,348 participants developed dementia (eTable 10). The mean baseline age was 64.6 years (SD = 8.9), 56.7% were female, and 47.1% had at least a college degree (Table 1). EHR encounter frequency differed by diabetes status, with participants classified as having T1DM having fewer individuals with no encounters 448 days (the 99th percentile of the distribution of days between encounters) after baseline, and a higher proportion with ≥3 encounters in the year following baseline compared with those without diabetes (eAppendix 1; eTables 11 and 12).

T1DM (vs no DM) was associated with much higher incidence of dementia, whether adjusted for sociodemographic factors (HR 2.82; 95% CI 2.28–3.48) or also for midlife factors (HR 2.38; 95% CI 1.84–3.08) (Table 2). T2DM was also associated with higher dementia risk, compared with not having DM (adjusted for sociodemographic factors, HR 2.08; 95% CI 1.87–2.31; additionally adjusted for midlife factors, HR 2.00; 95% CI 1.76–2.27).

Gender-Stratified Associations

T1DM was associated with higher hazard of dementia among women (set 1 HR 3.04; 95% CI 2.28–4.05; Figure 2) and men (set 1 HR 2.59; 95% CI 1.90–3.54). Among both women and men, T2DM was associated with approximately double the dementia hazard rate, compared with not having DM.

Race- and Ethnicity-Stratified Associations

T1DM was associated with elevated dementia incidence in all 3 race and ethnicity groups ($HR_{Hispanic/Latino}$ 3.29; 95% CI 2.02–5.35; HR_{White} 2.98; 95% CI 2.22–4.01; HR_{Other} 2.38; 95% CI 1.60–3.55; Figure 2). These associations changed little after adjustment for midlife factors (set 2).

In all racial and ethnic groups, T2DM was associated with higher dementia risk ($HR_{Hispanic/Latino}$ 2.94; 95% CI 2.16–4.01; HR_{White} 2.16; 95% CI 1.89–2.47; HR_{Other} 1.58; 95% CI 1.27–1.96; Figure 2). These associations remained after adjustment for midlife factors (Figure 2).

Attributable Fractions and Population Attributable Fractions

With a T1DM prevalence of 6.05% among individuals who developed dementia, we estimated that 3.90% of dementia cases among AoU participants and 64.5% of dementia cases among people with T1DM could be attributed to T1DM (eTable 13).

Sensitivity Analyses

Classifying Diabetes Type Defined Based on 3 or More EHR Encounters

Estimated associations of DM with dementia incidence were similar when we classified T1DM based on a stricter criterion-requiring more than 3 T1DM EHR encounters (eTable 14). Under this stricter definition, 2,685 participants were classified as having T1DM. Compared with individuals without DM, dementia incidence was higher among those with T1DM (HR 2.66; 95% CI 1.98–3.55) and among those with T2DM (HR 1.93; 95% CI 1.73–2.15), adjusting for sociodemographic factors. Gender-stratified results were also similar (women T1DM HR adjusted for sociodemographic factors 2.74; 95% CI 1.84–4.08; men T1DM HR 2.54; 95% CI 1.66–3.90; eTable 14). Owing to small sample sizes, we did not stratify by race and ethnicity.

Discussion

We developed an algorithm for identifying the optimal diabetes encounters to classify AoU participants with T1DM based on EHR encounters. Both T1DM and T2DM were associated with more than twice the risk of dementia compared with not having DM with moderately greater elevation in risk associated with T1DM. Estimates were similar across sex, and race and ethnicity.

A previous algorithm developed in the AoU cohort used EHR data, including laboratory results and prescription medications, as well as polygenic scores and self-reported diabetes status.[37] In this algorithm, insulin use was a criterion for

Table 1 Characteristics of the Analytic Sample, by Diabetes Status

	All (N = 283,772)	T1DM (n = 5,442)	T2DM (n = 51,511)	No diabetes (n = 226,819)
Age, y, mean (SD)	64.62 (8.96)	65.11 (8.72)	64.95 (8.85)	64.53 (8.99)
Woman, n (%)	160,813 (56.7)	2,741 (50.4)	27,775 (53.9)	130,297 (57.4)
Race and ethnicity, n (%)				
NH White	171,004 (60.3)	25,155 (48.8)	171,004 (60.3)	25,155 (48.8)
Hispanic/Latino	37,802 (13.3)	1,030 (18.9)	9,521 (18.5)	27,251 (12)
NH Other[a]	74,966 (26.4)	1,815 (33.4)	16,835 (32.7)	56,316 (24.8)
Educational attainment, n (%)				
Advanced degree	69,554 (24.5)	825 (15.2)	7,974 (15.5)	60,755 (26.8)
College graduate	64,103 (22.6)	1,059 (19.5)	9,433 (18.3)	53,611 (23.6)
Some college	72,129 (25.4)	1,590 (29.2)	15,128 (29.4)	55,411 (24.4)
High school or GED	46,639 (16.4)	1,073 (19.7)	10,906 (21.2)	34,660 (15.3)
Less than high school	24,062 (8.5)	739 (13.6)	6,659 (12.9)	16,664 (7.3)
BMI, kg/m^2, mean (SD)	29.83 (7.06)	31.99 (7.47)	32.87 (7.53)	28.83 (6.6)
Income (divided by the square root of household size), mean (SD)	57,248.69 (50,013.09)	40,387.87 (41,412.61)	41,986.11 (41,735.58)	60,808.77 (51,085.14)
Ever smoked (at least 100 Cigarettes in lifetime), n (%)	122,902 (43.3)	2,474 (45.5)	24,078 (46.7)	96,350 (42.5)
Alcohol use, n (%)				
Never	25,348 (8.9)	746 (13.7)	6,430 (12.5)	18,172 (8)
Past	54,150 (19.1)	1,638 (30.1)	13,445 (26.1)	39,067 (17.2)
Low	70,379 (24.8)	815 (15)	8,698 (16.9)	60,866 (26.8)
High	49,025 (17.3)	571 (10.5)	6,120 (11.9)	42,334 (18.7)

Abbreviations: BMI = body mass index; GED = general educational development; NH = non-Hispanic; T1DM = type 1 diabetes mellitus; T2DM = type 1 diabetes mellitus.
[a] The NH Other race and ethnicity category combines participants who identified as NH Asian, NH Black, NH Middle Eastern or North African, or other race categories. Sample sizes were too small to analyze these groups separately. Furthermore, the All of Us research program does not permit reporting results for cells with fewer than 20 participants.

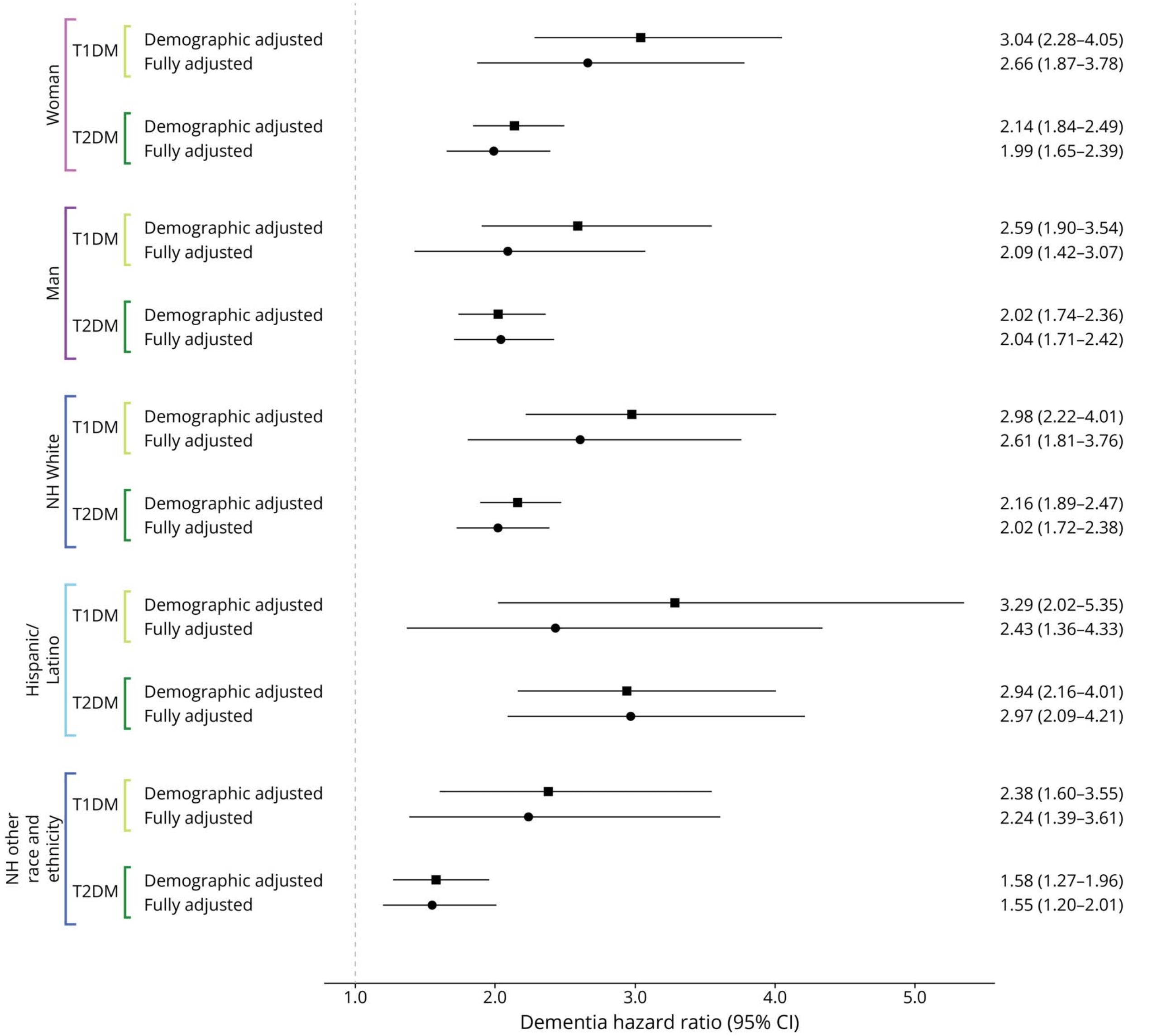
identifying individuals with T1DM, yet many individuals with T2DM use insulin in advanced stages of the disease.[38] Our approach builds on this work by validating DM type against 2 reference standards, including self-report and a clinical biomarker that provides a biologically grounded framework for distinguishing T1DM from T2DM. This distinction may be

Table 2 Dementia Hazard Ratio Associated With Each Type of Diabetes Mellitus (Reference: No Diabetes Mellitus), All Participants

		Dementia hazard ratio (95% CI), diabetes type (type 1 diabetes, type 2 diabetes, no diabetes [referent])	
	Dementia cases/total	**Adjusted for sociodemographic factors[a]**	**Adjusted for midlife factors[b]**
No diabetes (referent)	1,262/226,819	1.00	1.00
T1DM[c]	144/5,442	2.82 (2.28–3.48)	2.38 (1.84–3.08)
T2DM[c]	942/51,511	2.08 (1.87–2.31)	2.00 (1.76–2.27)

Abbreviations: T1DM = type 1 diabetes mellitus; T2DM = type 2 diabetes mellitus.
[a] Adjusted for sociodemographic factors: age, sex, racial and ethnicity, educational attainment, and household income.
[b] Fully adjusted models were further adjusted for the following midlife factors: smoking history and alcohol use.
[c] Reference group is no diabetes mellitus; type 1 diabetes includes individuals who have any form of T1DM (either T1DM alone or T1DM and T2DM); type 2 diabetes includes only individuals with T2DM alone.

Figure 2 Dementia Hazard Ratio Corresponding to Each Type of Diabetes Mellitus (Reference: No Diabetes Mellitus), Within Strata of Sex, and Race and Ethnicity



Hazard ratios and 95% CIs for incident dementia by diabetes type, stratified by sex and by race and ethnicity. Squares represent demographic-adjusted models, adjusted for age, sex, race and ethnicity, educational attainment, and household income. Circles represent fully adjusted models, which additionally account for midlife factors including smoking history and alcohol use. NH = non-Hispanic; T1DM = type 1 diabetes mellitus; T2DM = type 2 diabetes mellitus.

particularly important when evaluating outcomes, such as dementia, that may have etiologic pathways specific to DM type.

Our findings advance the existing evidence that DM is related to higher risk of dementia.[8-12] Despite the vast evidence amassed, there remains a need to evaluate dementia risk by DM type. For example, the 2024 Lancet Commission report on dementia prevention prioritizes diabetes as a modifiable risk factor for dementia, but this is primarily based on meta-analyses of studies evaluating T2DM or not specifying diabetes type (e.g., DM).[39] Thus, it remains important to understand the effect of T1DM on dementia risk, especially as life expectancy for individuals with T1DM lengthens. Our study offers valuable evidence by distinguishing individuals with T1DM from those with T2DM in a setting with a large number of individuals with T1DM.

We estimated that nearly two-thirds of incident dementia among adults with T1DM could be attributed to their diabetes status. T1DM is not common, so it accounts for a fairly small fraction of all dementia cases, but for the growing number of individuals with T1DM older than 65, these findings underscore the urgency of identifying and intervening on the mechanisms linking T1DM to dementia.[40,41]

Our findings regarding T2DM are consistent with previous literature. T2DM may contribute to dementia etiology through several mechanisms, including hyperglycemia, increased beta-amyloid deposition, or comorbidities such as metabolic syndrome, hyperinsulinemia, or stroke.[42-44] By contrast, T1DM is characterized by pancreatic beta-cell destruction and the need for insulin supplementation to address hypoglycemia. Hypoglycemic events may increase dementia risk through neuronal damage caused by altered glucose metabolism and insulin insufficiency, or through oxidative stress and inflammation in the hippocampus.[45-48] Although T2DM entails insulin resistance and hyperglycemia, T1DM may present a unique set of risks, in part due to the autoimmune destruction of beta cells. These mechanisms should be further explored.

With more than 5,000 participants identified with T1DM, our study is among the largest to examine the association between T1DM and dementia risk, although a previous study in England included more than 340,000 individuals with T1DM. Our analyses were feasible because we developed and validated an algorithm to accurately distinguish individuals with T1DM from those with T2DM. Our diabetes classification integrates multiple sources of information including diagnostic codes, laboratory measures, and self-report information, which could help capture the complexities of diabetes presentations and treatments, improving classification accuracy.

Our study has some limitations. Although our study followed participants from AoU enrollment to dementia incidence, it relies on preexisting EHR and survey data to classify DM status. This introduces potential limitations including reliance on health care systems to diagnose DM, the assumption that DM care occurred in settings where records were captured by AoU, participant knowledge of their DM status, nonuniform follow-up care, and, as a result, possible misclassification of DM. Similarly, our assessment of all-cause dementia relied on EHR-based diagnoses, which may have entailed misclassification of dementia status both at baseline and over follow-up. Dementia is often underdiagnosed and oftentimes delayed in routine clinical care. That said, individuals with T1DM may have more frequent contact with the health care system, which could increase their likelihood of earlier dementia diagnoses relative to those without DM. Second, although our algorithm for identifying T1DM demonstrated good sensitivity, even modest misclassification could bias our estimates given that T1DM is rare. For each reference standard, we selected a classification threshold that maximized the Youden J index to balance sensitivity and specificity; however, this threshold may not be optimal for all analytic purposes and may depend on the research question at hand. Misclassification of DM status could have resulted from incomplete information. For example, AoU participants without diabetes-related EHR encounters were classified as having no diabetes, but participants' EHR data may not capture care received from multiple networks. However, we believe that it was unlikely that individuals without DM were misclassified as having DM, particularly not T1DM. Third, C-peptide data, obtained from clinical laboratory records, resulted from routine care, rather than collected systematically from all AoU participants. As a result, individuals with these data may be more likely to have suspected or confirmed T1DM or more complete clinical records, while those with less frequent care or limited health care access may be underrepresented in these data. To address this, we also used self-reported diabetes information to identify individuals with T1DM who may not have corresponding laboratory data in the EHR. In addition, because we only used participants who self-reported their gender identity as man or woman, our analyses do not capture nonbinary or other gender identities, which limit generalizability to other populations. We also combined the small NH Asian, NH Black, and NH Middle Eastern and North African subgroups into a single NH Other race and ethnicity category, which may aggregate important subgroup differences and limit interpretation of race and ethnicity findings. Finally, AoU's recruitment process, largely convenience sampling through academic institutions and affiliated clinics, may limit generalizability. In a recent study comparing AoU with the nationally representative National Health and Nutrition Examination Survey, associations between various characteristics and all-cause mortality were in the same direction across the 2 studies, but some associations, especially those for clinical characteristics, were stronger in AoU.[49] These differences suggest that although our findings provide valuable insights, they may not generalize to the broader US population of adults aged 50 years and older.

This study provides strong evidence that both T1DM and T2DM are associated with an elevated risk of dementia, underscoring the need to consider diabetes subtype when evaluating long-term cognitive outcomes. By developing and validating a classification algorithm to distinguish T1DM from T2DM using EHR data, we addressed a critical methodologic need for more accurate surveillance of diabetes-related dementia risk in large population-based data sets. Our findings highlight the substantial burden of dementia among individuals with T1DM—particularly relevant because this population continues to age and suggests the need for targeted prevention strategies. Future research should explore the distinct biological mechanisms linking each diabetes subtype to dementia and quantify the effect of misclassification bias in studies relying on EHR-based diabetes phenotyping.

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Author Contributions

A.M. Pederson: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. P. Buto: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S.C. Zimmerman: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M. Velez: drafting/revision of the manuscript for content, including medical writing for content. K.D. Sims: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. A.R. Murchland: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J. Wang: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. A.T. Brennan: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M.M. Glymour: drafting/revision of the manuscript for content, including medical writing for content. J. Weuve: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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Disclosure

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